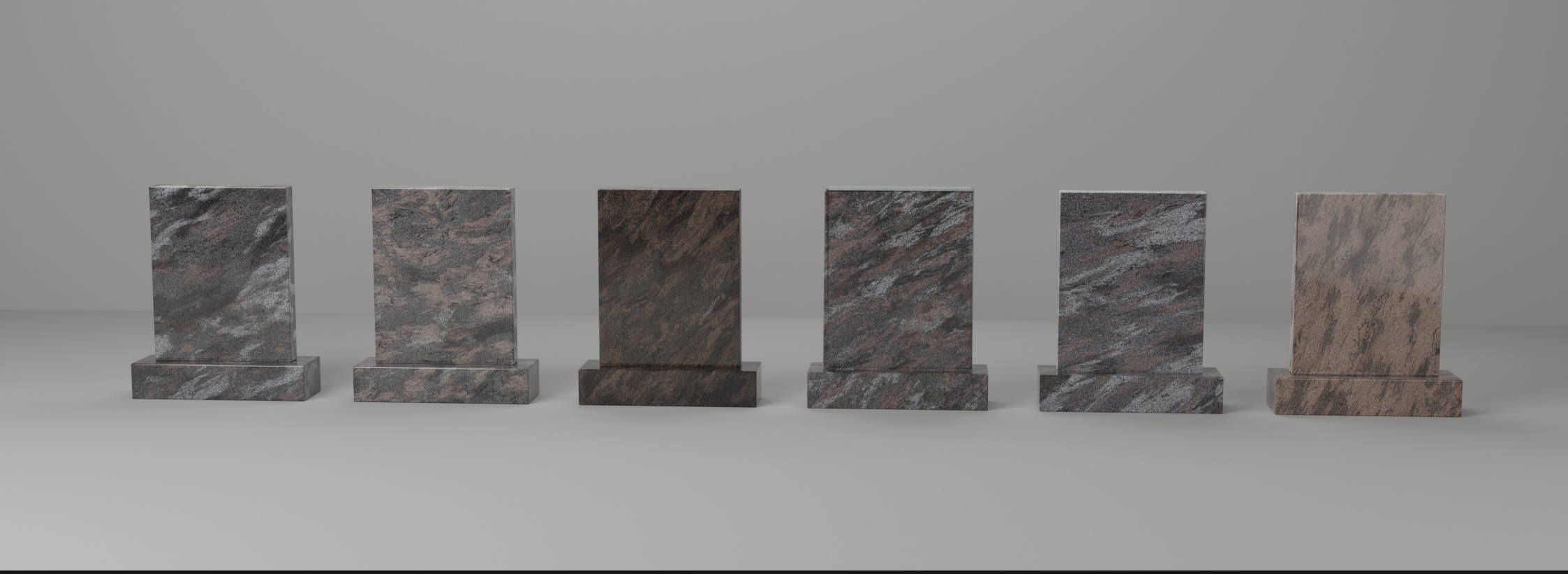


MATERIAL REPORT · SIX DATASETS

Procedural Headstone Granite

Seed-controlled Blender materials derived from photographs of six Paradiso-type gneiss memorials, calibrated against the photos and rendered in HD.

01
Salt & Pepper
folder 1st

02
Flame Swirl
folder 2nd

03
Dark Maroon
folder 3rd

04
Folded + Rock-pitched
folder 4th

05
High Contrast
folder 5th

06
Rust Red
folder 6th

Blender 5.2 LTS · Cycles · October 2026

OVERVIEW

Method and calibration

how the materials were built and checked

What the stone is

All six memorials are cut from the same family of migmatite gneiss sold as "Paradiso" or "Himalayan Blue" granite. Each material is built from the same seven geological layers, with per-dataset presets controlling colour, proportions and flow.

Matrix	fine ~1.5 mm crystals (Voronoi)
Melanosome	broad darker biotite-rich layers
Rose layers	garnet / K-feldspar clouds
Leucosome	grey quartz-feldspar clouds
Speckle	1–4 mm black biotite clots
Schlieren	thin broken black streaks
Lamination	fine foliation stripes along folds

Every layer is sampled in one folded flow space: rotate to the flow angle, stretch, then three levels of domain warping (sweep, fold, crinkle).

How it was checked

For each folder a 23–31 cm patch of clean polished stone was cut from a photo. The shader’s albedo was rendered at the same physical size with an orthographic camera, and the presets were tuned until colour, contrast, percentiles and redness matched.

Calibration: photo vs shader (same patch)

FOLDER	MEAN SRGB	CONTRAST Σ	P5–P95	REDDISH
01 1st	58 55 55	27.6	10–101	10%
photo / shader	**58 54 55**	**23.2**	**13–96**	**9%**
02 2nd	81 76 75	22.5	41–114	18%
photo / shader	**76 71 70**	**23.4**	**31–112**	**24%**
03 3rd	47 41 36	18.3	10–71	52%
photo / shader	**49 41 37**	**17.1**	**13–71**	**51%**
04 4th	59 57 56	25.7	13–98	9%
photo / shader	**55 52 51**	**23.5**	**12–93**	**12%**
05 5th	69 66 68	33.9	11–125	15%
photo / shader	**73 69 72**	**31.6**	**16–129**	**16%**
06 6th	82 69 62	27.1	27–116	94%
photo / shader	**84 71 65**	**24.3**	**31–113**	**93%**

Upper value in each pair is the photo, lower is the shader. Contrast σ is the luminance standard deviation (sRGB 0–255). The first version had σ ≈ 7 and periodic banding; v2 fixes both.

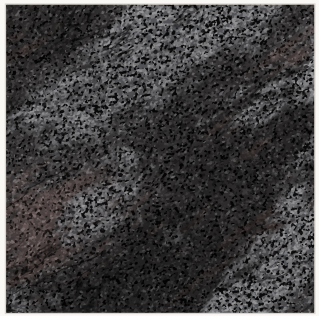
OVERVIEW

Photo vs shader at a glance

each pair covers the same 23–31 cm of stone

01 Salt & Pepper

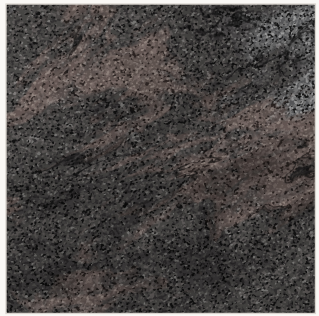
PHOTO



SHADER

02 Flame Swirl

PHOTO



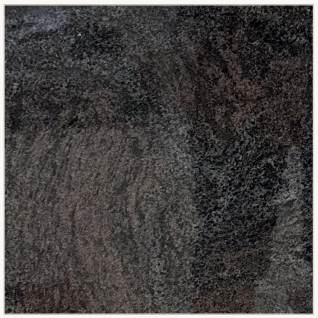
SHADER

03 Dark Maroon

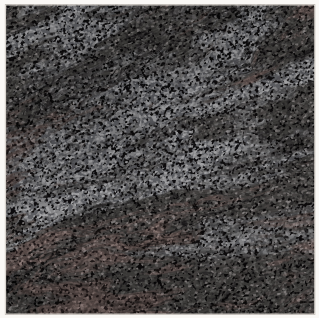
PHOTO

SHADER

04 Folded + Rock-pitched



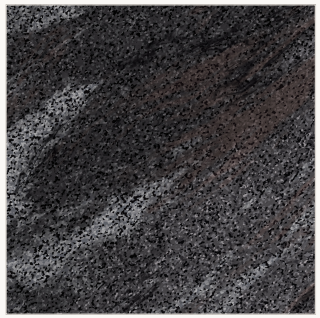
PHOTO



SHADER

05 High Contrast

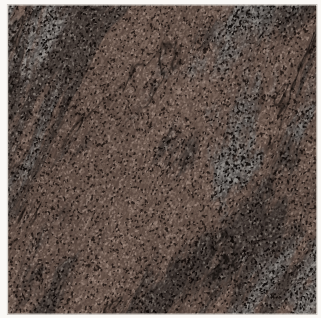
PHOTO



SHADER

06 Rust Red

PHOTO



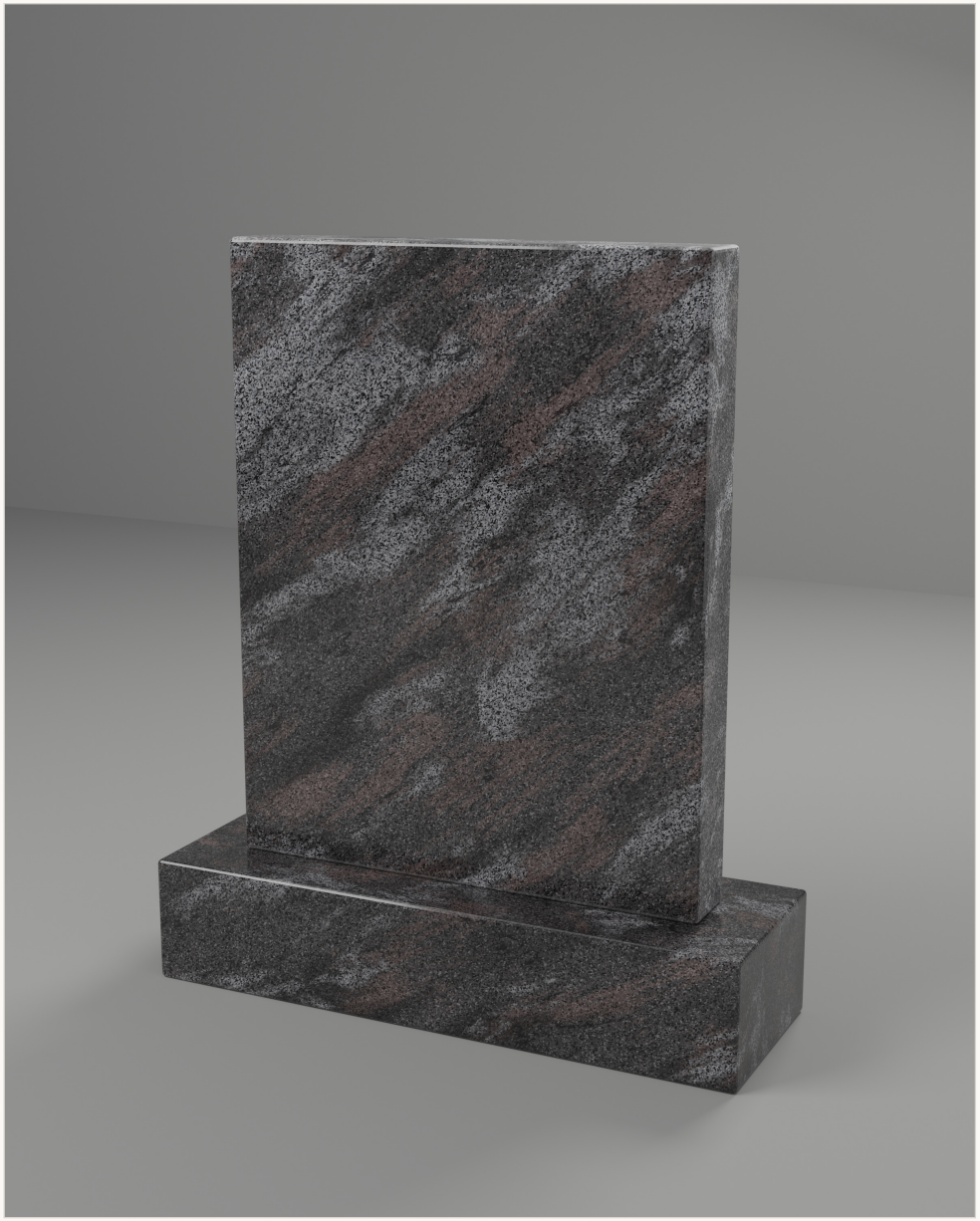
SHADER

Photo patches are cropped from the original 12 MP photographs at the exact calibration area; shader patches are 2048 px albedo renders of the same physical size. The shader has no lighting, reflections or lens blur, so it reads slightly crisper than the photos. Full-resolution versions of every pair appear on each dataset’s surface-detail page.

Salt & Pepper

GS01_SaltPepper · seed 1

REFERENCE PHOTOGRAPH folder 1st · original 12 MP



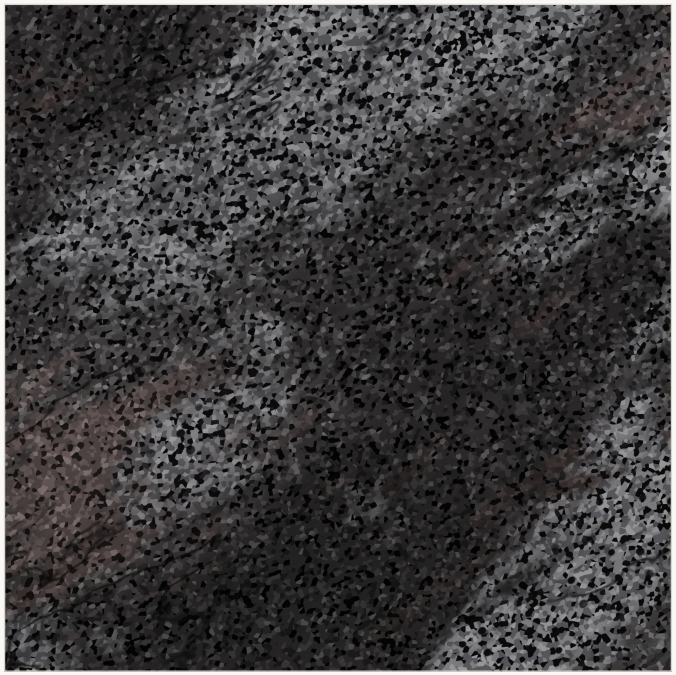
PROCEDURAL MATERIAL Cycles render · 2400 × 3000 · polished + honed

DATASET 01 · FOLDER 1st · REF 269567

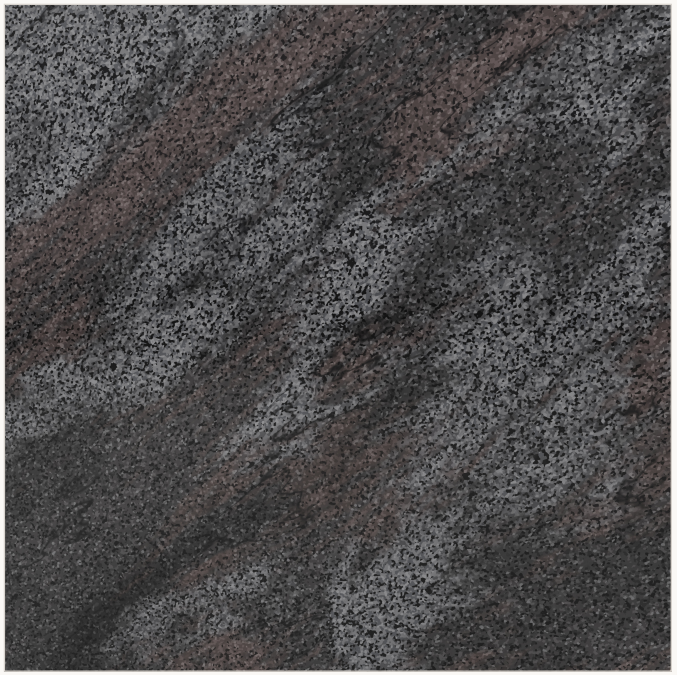
Salt & Pepper — surface detail

same physical patch, then lit close-up

PHOTO PATCH ~23 cm wide



SHADER ALBEDO 2048 px · same area



LIT CLOSE-UP 2400 px · polished face

Observed in the photographs

- Neutral charcoal base with the most grey "salt-and-pepper" clouds of the six.
- Dusty-rose layers are weak; black biotite clots are large and frequent.
- Flow rises ~32° to the right with gentle sweeps.

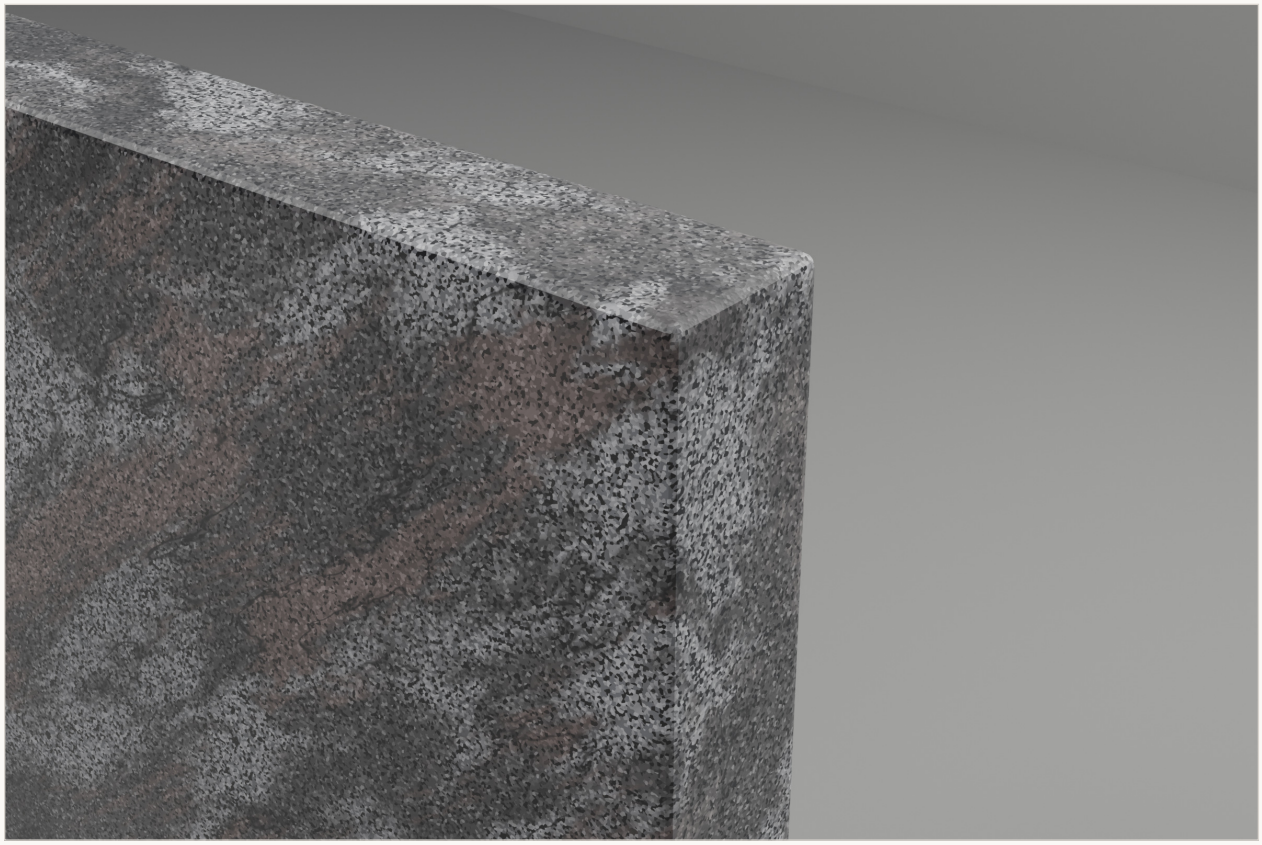
Calibration (photo → shader)

Mean sRGB	58 55 55	→	**58 54 55**
Contrast σ	27.6	→	**23.2**
p5–p95	10–101	→	**13–96**
Reddish pixels	10%	→	**9%**

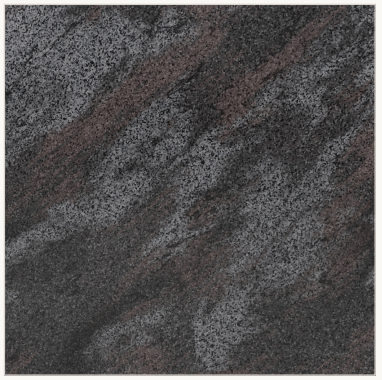
DATASET 01 · FOLDER 1st · REF 269567

Salt & Pepper — finish and seed variation

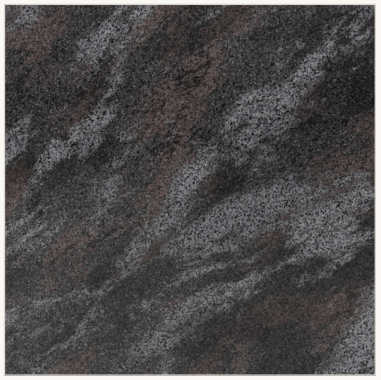
edge close-up · same material, four seeds



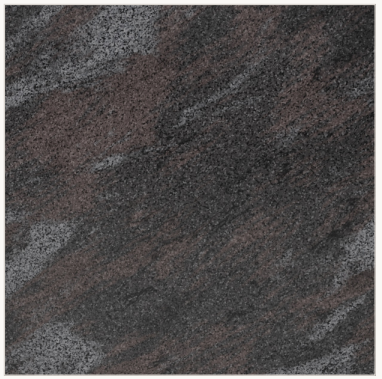
HONED EDGE polished face meets the side / top finish · 2400 × 1600



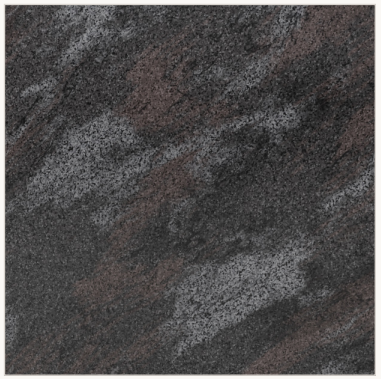
SEED 1 (DEFAULT)



SEED 11



SEED 222



SEED 3333

Finish

The wave-cut side panels are honed: the pattern stays visible but lifts to a light, cool grey.

Key preset values

scale	**9**	flow angle	**32°**	stretch	**0.35**	warp	**0.9**
fold	**0.35**	rose	**0.38**	leucosome	**0.7**	melanosome	**0.45**
biotite	**0.6**	speckle	**0.6**	lamination	**0.1**	saturation	**0.7**